

Exercises sheet 3: Graph Theory

Exercise 1. For each of the sequences indicated below, decide if it represents the degree sequence of a simple graph:

1. 1, 2, 3, 3, 5
2. 2, 3, 4, 5, 3, 4
3. 2, 2, 2
4. 5, 3, 5, 1, 2, 2

Exercise 2. Show that the number of K_3 subgraphs in a complete graph K_n with $n \geq 3$ is equal to $\frac{n(n-1)(n-2)}{6}$.

Exercise 3. Let G be a graph and the set of vertices $V = \{2, 3, 4, 5, 11, 12, 13, 14\}$, and two vertices u and v are adjacent if $\gcd(u, v) = 1$.

1. Draw G .
2. Find $e(G)$, $\delta(G)$, $\Delta(G)$, $N(4)$, $N[3]$ and $d(13)$

Exercise 4. Does there exist a simple graph of order 5, such that there a vertex of degree $\Delta(G) = 4$ and a vertex of degree $\delta(G) = 1$? What about two vertices of degree $\Delta(G) = 4$ and a vertex with degree $\delta(G) = 1$?

Exercise 5. Let G be a simple graph. Show that if $P = v_1 \dots v_l$ is a longest path in G , then all the neighbors of v_1 are in $\{v_2, \dots, v_l\}$.

Exercise 6. Let the set of vertices $V = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$, and two vertices u and v are adjacent if $|u - v| \in \{1, 4, 5, 8\}$. Determine the order and the size of the following subgraphs of G :

1. The subgraph induced by even vertices.
2. The subgraph induced by odd vertices.
3. The subgraph induced by the set $\{0, 1, 2, 3, 4\}$.
4. A spanning subgraph with as many edges as possible but without cycles.

Exercise 7. Let u and v be two vertices of a graph G . Show that the following conditions are equivalent:

- (a) There exists a walk between u and v in G .
- (b) There exists a simple path between u and v in G . (A simple path is a path with no repeated vertices.)

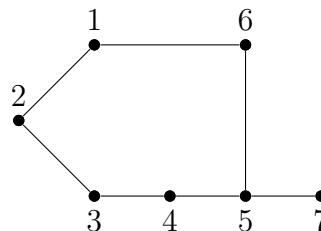
(c) There exists an elementary path between u and v in G . (An elementary path is a path with no repeated edges.)

Exercise 8. How many distinct simple graphs can be defined on n distinct given vertices?

Exercise 9. Show that the graph obtained from a tree, by deleting or adding a leaf (pendant vertex), is a tree.

Exercise 10. Let G be a graph. Specify the following characteristics of this graph:

1. $\omega(G)$ (Clique number)
2. $\chi(G)$ (Chromatic number)
3. $\alpha(G)$ (Independence number)
4. Non-connected subgraph
5. Connected spanning subgraph.



Exercise 11. Can a bipartite graph contain an odd-length cycle? Give an example or justify your answer.

Exercise 12. Let G be a tree with at least two vertices, and let v be a vertex of G . Show that the number of connected components of $G - v$ is equal to $d(v)$.

Exercise 13. Show that the graph obtained from a tree, by deleting or adding a leaf (pendant vertex), is a tree.

Exercise 14. Which complete graphs K_n admit an Eulerian cycle?

Exercise 15. What does a graph with chromatic number 1 look like? What is the chromatic number of a tree?

Exercise 16. Show that any acyclic graph of order n has at most $n - 1$ edges.